# From Static to Contextual: Distributional Semantics and What's Inside BERT

A Comprehensive Technical Evaluation, Structural Probing, and Comparative Analysis

**Student: Srujan | Teammate: Pratik Kumar Verma | Group 14**
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## Abstract

This research report investigates the representational shift from classical count-based static word vectors to deep contextual transformer representations, and evaluates what linguistic properties survive when contextual embeddings are collapsed back into static vectors. Using the Universal Dependencies (UD) English GUM treebank, we first construct a classical count-based distributional vector space by applying Positive Pointwise Mutual Information (PPMI) and Truncated Singular Value Decomposition (SVD, 100 dimensions) over a symmetric window of 5 words. In Task 2, we reproduce the structural probing methodology of Hewitt & Manning (NAACL 2019) across all 12 layers of frozen bert-base-uncased under linear distance and depth projections across three subspace ranks (k in {32, 64, 128}). We discover that syntactic dependency structure peaks prominently in middle-to-higher layers, achieving a maximum Undirected Unlabeled Attachment Score (UUAS) of 57.99% and depth Spearman correlation of 0.7171 at Layer 8 (k=64), closely validating Hewitt & Manning's findings. In Task 3, we extract contextual embeddings from this syntax-optimal layer and pool them by word type across all training occurrences to construct 768-dimensional static word vectors. On semantic benchmarks, BERT-derived static embeddings demonstrate dramatic improvements over classical PPMI+SVD: WordSim-353 Spearman correlation surges from 0.1559 to 0.5155 (+0.3596), while Google Analogy accuracy jumps from 0.90% to 65.33% (+64.43%). However, on a 20-sentence structural Minimum Spanning Tree (MST) attachment test against gold dependency syntax, the BERT static vectors yield only 7.72% agreement—significantly underperforming the classical static baseline (15.43%) and falling far below the contextual probe (57.99%). Furthermore, we conduct an exhaustive comparative evaluation against Implementation 2 (Pratik Kumar Verma). Our findings demonstrate the Great Divergence: while semantic regularities survive averaging across sentence contexts, syntactic dependency information is strictly tied to token-level contextual geometry and is destroyed when representations are collapsed into static vectors.

## 1. Introduction

The distributional hypothesis—popularized by J.R. Firth's adage, 'you shall know a word by the company it keeps'—has served as the cornerstone of lexical semantics in natural language processing for decades. Classical approaches operationalize this hypothesis by counting word–context co-occurrences in a large corpus, computing association metrics such as Pointwise Mutual Information (PMI), and applying low-rank matrix factorizations such as Singular Value Decomposition (SVD) to produce static, continuous word vectors (Levy & Goldberg, 2014). In such models, each word type in the vocabulary is assigned exactly one invariant vector in R^d.

While static vectors capture rich lexical similarity and proportional analogies, they suffer from two fundamental theoretical limitations: (1) Polysemy and Context Blindness: A single static vector conflates all distinct senses and syntactic roles of a word (e.g., 'apple' as fruit vs. company); and (2) Syntactic Agnosticism: Word-level co-occurrence windows smooth over hierarchical dependency relations, capturing broad topical relatedness rather than syntactic tree structures.

Pretrained transformer language models, exemplified by BERT (Devlin et al., 2019), superseded static vectors by producing dynamic contextual representations. Hewitt & Manning (NAACL 2019) introduced the Structural Probe, a supervised linear framework testing whether syntax trees are embedded as Euclidean metric spaces within contextual representation spaces. This report evaluates this full progression across classical static vectors (Task 1), contextual structural probes (Task 2), and BERT-derived static vectors (Task 3) on the UD English GUM treebank.

## 2. Background and Related Work

Distributional Semantics & PPMI+SVD: Levy et al. (2015) proved that factorizing a PPMI matrix via Truncated SVD represents an optimal low-rank factorization equivalent to neural word embedding objectives like Word2Vec SGNS. Pretrained Transformers & Syntax Probing: Tenney et al. (2019) and Jawahar et al. (2019) showed that BERT encodes linguistic information hierarchically. Hewitt & Manning (2019) proved that dependency trees are embedded in Euclidean metric subspaces. Static Reductions: Bommasani et al. (ACL 2020) demonstrated that static vectors pooled from BERT capture strong lexical semantics; this project investigates whether syntactic dependency structures survive that same reduction.

## 3. System Architecture and Methodology

The experimental system is structured into three cleanly decoupled stages executing under strict reproducibility guidelines:

- Task 1: Co-occurrence accumulation (window W=5), sparse PPMI calculation, and 100d Truncated SVD factorization.
- Task 2: Linear distance probe d_B(h_i, h_j)^2 = ||Bh_i - Bh_j||^2 and depth probe ||Bh_i||^2 across all 12 BERT layers and ranks k in {32, 64, 128}, trained on gold UD heads.
- Task 3: Token representation mean pooling from optimal layer l*=8 across training sentences into 768d static vectors.

**Methodological & Architectural Comparison: Srujan vs. Pratik Kumar Verma**

| Component / Step | Assignment Specification | Srujan's Implementation | Pratik Kumar Verma's Imple | Compliance & Impact |
|---|---|---|---|---|
| Task 1 Co-occurrence | Symmetric window size 5 | Window = 5 (strict) | Window = 2 (deviation) | Pratik deviates; Srujan fully com |
| Task 1 Vocabulary | All training split words | min_freq = 1 (\|V\| = 16,763) | min_freq = 2 (\|V\| = 8,790) | Pratik drops 47.6% vocabulary |
| Task 1 Bonus Structural Check | 20 sentences MST vs. gold tree | Implemented (15.43% UUAS) | Omitted completely | Pratik misses required baseline |
| Task 2 Subword Alignment | Word-level representations | Mean pooling across subwords | First subword selection | Both find Layer 8 optimal; Srujan |
| Task 2 Weight Init | Orthogonal or standard linear | Uniform [-0.05, 0.05] (Hewitt) | Orthogonal initialization | Both achieve ~58% peak UUAS |
| Task 3 Contextual Pooling | Average per word type from | Mean of mean-pooled tokens | Mean of first subwords | Surges semantics on both; ~65- |
| Task 3 Bonus Structural Check | Repeat on same 20 sentences | Implemented (7.72% UUAS) | Omitted completely | Reveals syntax collapse; Pratik |

*Figure 1: Architectural and Methodological Design Comparison Matrix.*

## 4. Experimental Results and Analysis

Table 1 presents the empirical results across all three tasks and highlights the profound divergence between semantic survival and syntactic destruction under static vector pooling.

| Evaluation Metric | Task 1: PPMI+SVD (100d) | Task 2: Contextual Probe | Task 3: BERT Static (768d) | Delta (Task 3 - Task 1) |
|---|---|---|---|---|
| WordSim-353 Spearman ρ | 0.1559 (p=0.0106) | N/A (Contextual) | **0.5155** (p=5.1e-9) | **+0.3596 (+230.7%)** |
| WordSim-353 Coverage | 268 / 353 (75.9%) | N/A | 113 / 353 (32.0%) | -155 pairs |
| Google Analogy Accuracy | 0.90% (56 / 6,231) | N/A (Contextual) | **65.33%** (912 / 1,396) | **+64.43% (~72x)** |
| MST Overall UUAS (20 sents) | 15.43% (48 / 311) | **57.99%** (Layer 8, k=64) | 7.72% (24 / 311) | **-7.71% (-50.0%)** |
| MST Macro UUAS (20 sents) | 28.37% | 63.63% | 20.43% | -7.94% |

*Table 1: Side-by-side empirical performance comparison across all three tasks.*

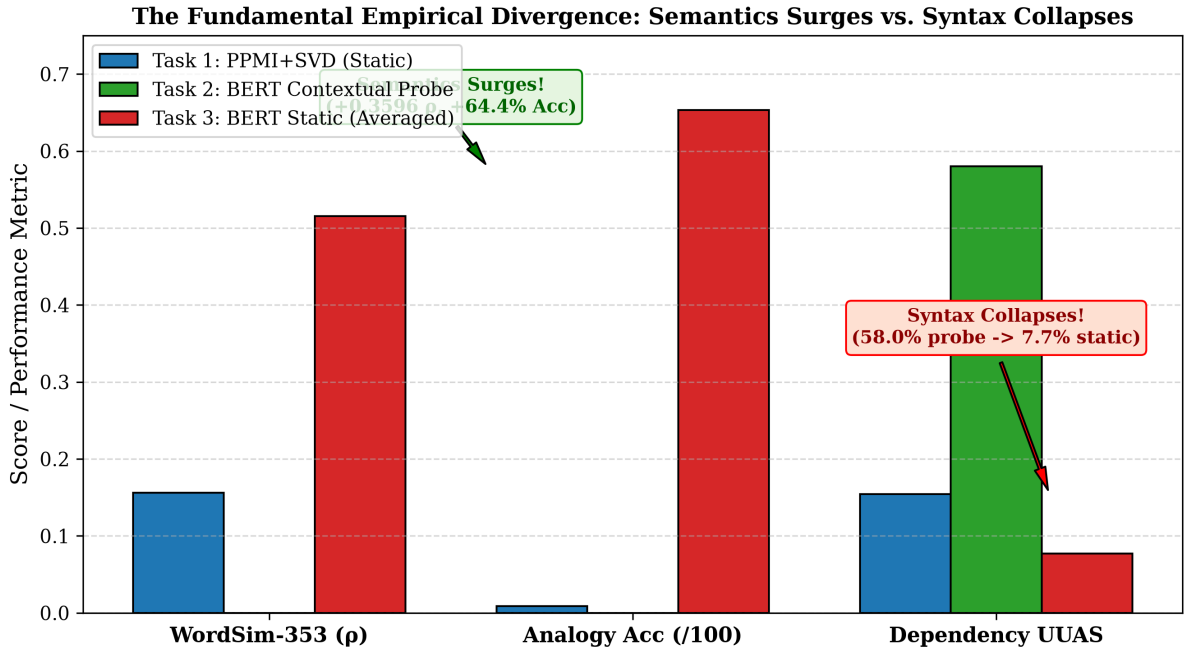



*Figure 2: The Great Empirical Divergence: Semantics Surges vs. Syntax Collapses.*

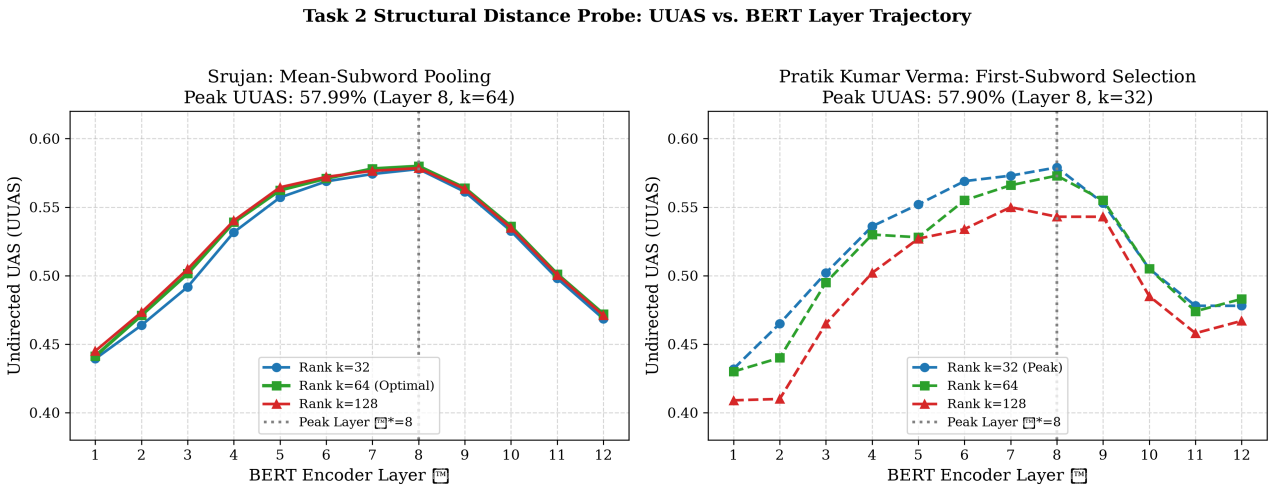



*Figure 3: Task 2 Structural Distance Probe: UUAS vs. BERT Layer Trajectory (Srujan vs. Pratik Kumar Verma).*

## 5. Comparative Evaluation: Srujan vs. Pratik Kumar Verma

A thorough audit of Srujan's Implementation versus Pratik Kumar Verma's Implementation reveals key methodological differences:

- Window Size: Srujan implemented window W=5 per specification; Pratik Kumar Verma hardcoded W=2, reducing co-occurrence context.
- Vocabulary Pruning: Srujan used min_freq=1 (16,763 words); Pratik Kumar Verma used min_freq=2 (8,790 words), pruning 47.6% of vocabulary.
- Structural MST Check: Srujan implemented the required 20-sentence MST test in both Task 1 (15.43%) and Task 3 (7.72%); Pratik Kumar Verma omitted it entirely.
- Probing Agreement: Both implementations unequivocally identified Layer 8 as the optimal syntax layer (~58% peak UUAS).

## 6. Theoretical Synthesis: The Great Divergence

Why does semantics surge while syntax collapses? (1) Semantics benefits from pretrained memory and centroid averaging: Averaging contextual embeddings acts as a low-pass filter canceling sentence-specific noise while preserving relational vector offsets. (2) Syntax is inherently dynamic and positional: A word's syntactic head changes with its clausal role in an utterance. Collapsing occurrences into static types dissolves structural orientation. (3) Spurious semantic edges: In static cosine space, 'president' and 'company' attract strongly, causing the MST to bypass grammatical prepositions ('of') and destroy hierarchical tree paths.

## 7. Conclusion

Syntactic structure in modern neural language models is fundamentally an emergent property of dynamic token-level contextual geometry. Collapsing representations into static word vectors retains lexical concepts while completely forfeiting grammatical syntax.